

1 Question 1

Let A be defined as:

$$A := \begin{bmatrix} 4 & 0 & 1 \\ 0 & 2 & 0 \\ 1 & 0 & 4 \end{bmatrix}$$

This matrix is symmetric. As a hint, $\lambda = 2$ is one eigenvalue.

- Find all eigenvalues of A .
- Find an eigenvector for each eigenvalue.
- Verify that your eigenvectors are mutually orthogonal.
- Normalize your eigenvectors to form an orthogonal matrix Q , and write down a diagonal matrix Λ such that

$$A = Q\Lambda Q^T.$$

1.1 Find all eigenvalues of A .

$$A - \lambda I := \begin{bmatrix} 4 - \lambda & 0 & 1 \\ 0 & 2 - \lambda & 0 \\ 1 & 0 & 4 - \lambda \end{bmatrix}$$

Determinant:

$$(2 - \lambda) \det \begin{bmatrix} 4 - \lambda & 1 \\ 1 & 4 - \lambda \end{bmatrix}$$

Characteristic Equation:

$$\begin{aligned} 0 &= (2 - \lambda) [(4 - \lambda)^2 - 1] \\ 0 &= (2 - \lambda)(4 - \lambda)^2 - (2 - \lambda) \\ 0 &= (2 - \lambda)(16 - 8\lambda + \lambda^2) - (2 - \lambda) \\ 0 &= (2 - \lambda)[(16 - 8\lambda + \lambda^2) - 1] \\ 0 &= (2 - \lambda)[15 - 8\lambda + \lambda^2] \\ 0 &= (2 - \lambda)(\lambda - 3)(\lambda - 5) \\ \therefore \lambda_2 &= 2 \quad \lambda_3 = 3 \quad \lambda_5 = 5 \end{aligned}$$

1.2 Find an eigenvector for each eigenvalue.

Let $M_n := A - \lambda_n I$. $n \in \{2, 3, 5\}$ Let $V_n := \mathcal{B}[Nul(M_n)]$. $n \in \{2, 3, 5\}$

$$A - \lambda_2 I = \begin{bmatrix} 2 & 0 & 1 \\ 0 & 0 & 0 \\ 1 & 0 & 2 \end{bmatrix} \implies Nul(A - \lambda_2 I) = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix} \implies \begin{cases} x_2 = 0 \\ x_2 = 1 \\ x_2 = 0 \end{cases} \implies V_1 = \left\{ \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} \right\}$$

$$A - \lambda_3 I := \begin{bmatrix} 1 & 0 & 1 \\ 0 & -1 & 0 \\ 1 & 0 & 1 \end{bmatrix} \implies \text{Nul}(A - \lambda_3 I) = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix} \implies \begin{cases} x_3 = -1 \\ x_3 = 0 \\ x_3 = 1 \end{cases} \implies V_3 = \left\{ \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix} \right\}$$

$$A - \lambda_5 I := \begin{bmatrix} -1 & 0 & 1 \\ 0 & -3 & 0 \\ 1 & 0 & -1 \end{bmatrix} \text{Nul}(A - \lambda_5 I) = \begin{bmatrix} 1 & 0 & -1 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix} \implies \begin{cases} x_3 = 1 \\ x_3 = 0 \\ x_3 = 1 \end{cases} \implies V_5 = \left\{ \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} \right\}$$

$$V_1 = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} \quad V_3 = \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix} \quad V_5 = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}$$

1.3 Verify that your eigenvectors are mutually orthogonal.

$$V_1^T V_3 := [0 \ 1 \ 0] \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix} = 0 \quad V_1^T V_5 [0 \ 1 \ 0] \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} = 0 \quad V_3^T V_5 [-1 \ 0 \ 1] \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} = 0$$

1.4 Normalize your eigenvectors to form an orthogonal matrix Q , and write down a diagonal matrix Λ such that $A = Q\Lambda Q^T$

$$V_1 = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} \implies q_1 = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} \quad V_3 = \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix} \implies q_3 = \begin{bmatrix} -\frac{1}{\sqrt{2}} \\ 0 \\ \frac{1}{\sqrt{2}} \end{bmatrix} \quad V_5 = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} \implies q_5 = \begin{bmatrix} \frac{1}{\sqrt{2}} \\ 0 \\ \frac{1}{\sqrt{2}} \end{bmatrix}$$

$$A = \begin{bmatrix} 4 & 0 & 1 \\ 0 & 2 & 0 \\ 1 & 0 & 4 \end{bmatrix} \quad Q = \begin{bmatrix} 0 & -\frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ 1 & 0 & 0 \\ 0 & \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \end{bmatrix} \quad \Lambda = \begin{bmatrix} 2 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 5 \end{bmatrix} \quad Q^T = \begin{bmatrix} 0 & 1 & 0 \\ -\frac{1}{\sqrt{2}} & 0 & \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & 0 & \frac{1}{\sqrt{2}} \end{bmatrix}$$

Final expression:

$$\begin{bmatrix} 4 & 0 & 1 \\ 0 & 2 & 0 \\ 1 & 0 & 4 \end{bmatrix} = \begin{bmatrix} 0 & -\frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ 1 & 0 & 0 \\ 0 & \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \end{bmatrix} \begin{bmatrix} 2 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 5 \end{bmatrix} \begin{bmatrix} 0 & 1 & 0 \\ -\frac{1}{\sqrt{2}} & 0 & \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & 0 & \frac{1}{\sqrt{2}} \end{bmatrix}$$

2 Question 2

Let A be defined as:

$$A = \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 2 \end{bmatrix}$$

(a) Find all eigenvalues of A and their algebraic multiplicities.

- (b) Find the eigenspace corresponding to each eigenvalue.
 (c) Find all eigenpairs of A .
 (d) Decide whether A is diagonalizable. Explain briefly.

$$A - \lambda I = \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 2 \end{bmatrix}$$

2.1 Find all eigenvalues of A and their algebraic multiplicities.

$$A - \lambda I = \begin{bmatrix} 1 - \lambda & 1 & 0 \\ 0 & 1 - \lambda & 0 \\ 0 & 0 & 2 - \lambda \end{bmatrix}$$

$$\det(A - \lambda I) = \begin{vmatrix} 1 - \lambda & 1 & 0 \\ 0 & 1 - \lambda & 0 \\ 0 & 0 & 2 - \lambda \end{vmatrix}$$

Determinant:

$$0 = \det(A - \lambda I) = (1 - \lambda) \begin{vmatrix} 1 - \lambda & 0 \\ 0 & 2 - \lambda \end{vmatrix} - 1 \begin{vmatrix} 0 & 0 \\ 0 & 2 - \lambda \end{vmatrix} + 0 \begin{vmatrix} 0 & 1 - \lambda \\ 0 & 0 \end{vmatrix}$$

$$0 = (1 - \lambda)[(1 - \lambda)(2 - \lambda) - 0] - 0$$

$$0 = (1 - \lambda)^2(2 - \lambda)$$

Characteristic Equation:

$$(1 - \lambda)^2(2 - \lambda) = 0$$

$$\implies \lambda_1 = 1 \quad (\text{Algebraic Multiplicity} = 2)$$

$$\implies \lambda_2 = 2 \quad (\text{Algebraic Multiplicity} = 1)$$

2.2 Find the eigenspace corresponding to each eigenvalue.

$$A - \lambda I = \begin{bmatrix} 1 - \lambda & 1 & 0 \\ 0 & 1 - \lambda & 0 \\ 0 & 0 & 2 - \lambda \end{bmatrix}$$

$$\implies \lambda_1 = 1 \quad (\text{Algebraic Multiplicity} = 2)$$

$$\implies \lambda_2 = 2 \quad (\text{Algebraic Multiplicity} = 1)$$

2.3 Find the eigenspace corresponding to each eigenvalue.

$$\text{Let } M_n := A - \lambda_n I \text{ . } n \in \{2, 3, 5\} \quad \text{Let } V_n := \mathcal{B}[\text{Nul}(M_n)] \text{ . } n \in \{2, 3, 5\}$$

$$A - \lambda I = \begin{bmatrix} 1 - \lambda & 1 & 0 \\ 0 & 1 - \lambda & 0 \\ 0 & 0 & 2 - \lambda \end{bmatrix}$$

$$A - \lambda_1 I = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix} \text{Nul}(A - \lambda_1 I) = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix} \implies \begin{cases} x_1 = 1 \\ x_1 = 0 \\ x_1 = 0 \end{cases} \implies V_1 = \left\{ \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} \right\}$$

$$A - \lambda_2 I = \begin{bmatrix} -1 & 1 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 0 \end{bmatrix} \implies \text{Nul}(A - \lambda_2 I) = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix} \implies \begin{cases} x_3 = 0 \\ x_3 = 0 \\ x_3 = 1 \end{cases} \implies V_2 = \left\{ \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \right\}$$

2.4 Find all eigenpairs of A .

$$(\lambda_1, V_1) \quad (\lambda_2, V_2)$$

2.5 Decide whether A is diagonalizable. Explain briefly.

Geometric Multiplicity < Algebraic Multiplicity $\implies A$ is NOT Diagonalizable

3 Question 3

Let A be defined as:

$$A = \begin{bmatrix} 1 & -2 \\ 2 & 1 \end{bmatrix}$$

- Find the characteristic polynomial of A .
- Find the eigenvalues of A over $\mathbb{C}$.
- For each eigenvalue, find an eigenvector in $\mathbb{C}^2$.
- Explain why A has no real eigenvectors.

3.1 Find the characteristic polynomial of A .

$$A - \lambda I = \begin{bmatrix} 1 - \lambda & -2 \\ 2 & 1 - \lambda \end{bmatrix}$$

Determinant:

$$\det(A - \lambda I) = \begin{vmatrix} 1 - \lambda & -2 \\ 2 & 1 - \lambda \end{vmatrix} = (1 - \lambda)(1 - \lambda) - (-2) \cdot 2 = (1 - \lambda)^2 + 4$$

3.2 Find the eigenvalues of A over $\mathbb{C}$.

$$\begin{aligned} (1 - \lambda)^2 + 4 &= 0 \\ 1 - 2\lambda + \lambda^2 + 4 &= 0 \\ 5 - 2\lambda + \lambda^2 &= 0 \end{aligned}$$

Therefore there must exist roots $\in \mathbb{C}$. Obtain complex conjugate:

$$\text{Quadratic Formula: } \lambda = \frac{2 \pm \sqrt{(-2)^2 - 4 \cdot 1 \cdot 5}}{2 \cdot 1} = 1 \pm 2i$$

Thus we obtain the following:

$$\lambda_1 = 1 + 2i \quad \lambda_2 = 1 - 2i$$

3.3 For each eigenvalue, find an eigenvector in $\mathbb{C}^2$.

$$\text{Let } M_n := A - \lambda_n I \text{ . } n \in \{2, 3, 5\} \quad \text{Let } V_n := \mathcal{B}[\text{Nul}(M_n)] \text{ . } n \in \{2, 3, 5\}$$

For $\lambda_1 = 1 + 2i$:

$$M_1 = A - (1 + 2i)I = \begin{bmatrix} 1 - (1 + 2i) & -2 \\ 2 & 1 - (1 + 2i) \end{bmatrix} = \begin{bmatrix} -2i & -2 \\ 2 & -2i \end{bmatrix}$$

To find the null space, we solve $M_1 \mathbf{v} = 0$:

$$-2ix - 2y = 0 \implies 2y = -2ix \implies y = -ix$$

Choosing $x = 1$, we get $y = -i$.

$$V_1 = \left\{ \begin{bmatrix} 1 \\ -i \end{bmatrix} \right\}$$

For $\lambda_2 = 1 - 2i$:

$$M_2 = A - (1 - 2i)I = \begin{bmatrix} 1 - (1 - 2i) & -2 \\ 2 & 1 - (1 - 2i) \end{bmatrix} = \begin{bmatrix} 2i & -2 \\ 2 & 2i \end{bmatrix}$$

To find the null space, we solve $M_2 \mathbf{v} = 0$:

$$2ix - 2y = 0 \implies 2y = 2ix \implies y = ix$$

Choosing $x = 1$, we get $y = i$.

$$V_2 = \left\{ \begin{bmatrix} 1 \\ i \end{bmatrix} \right\}$$

3.4 Explain why A has no real eigenvectors.

A matrix A has a real eigenvector $\mathbf{v} \in \mathbb{R}^2$ if and only if there exists a real eigenvalue $\lambda \in \mathbb{R}$ such that $A\mathbf{v} = \lambda\mathbf{v}$.

If we assume $\mathbf{v}$ is a real, non-zero vector, then the product $A\mathbf{v}$ must result in another real vector. For the equation $A\mathbf{v} = \lambda\mathbf{v}$ to hold, λ would necessarily have to be a real number. Because the eigenvalues of this matrix are strictly complex ($\lambda = 1 \pm 2i$), there is no real scalar λ that can satisfy the eigenvalue equation for any non-zero real vector $\mathbf{v}$.

Therefore, any eigenvector of A must necessarily have complex components, placing it in $\mathbb{C}^2 \setminus \mathbb{R}^2$.

4 Question 4

Find one singular value decomposition of the 3×2 matrix. Let A be defined as:

$$A = \begin{bmatrix} 1 & 1 \\ 1 & 1 \\ 0 & 0 \end{bmatrix}.$$

More precisely:

- (a) Compute $A^T A$ and find its eigenvalues.

- (b) Find the singular values of A .
- (c) Find an orthogonal matrix V whose columns are right singular vectors.
- (d) Find an orthogonal matrix U and a 3×2 diagonal matrix Σ such that

$$A = U\Sigma V^T.$$

4.1 Compute $A^T A$ and find its eigenvalues.

First, we find the transpose of A :

$$A^T = \begin{bmatrix} 1 & 1 & 0 \\ 1 & 1 & 0 \end{bmatrix}$$

Now, compute the product $A^T A$:

$$A^T A = \begin{bmatrix} 1 & 1 & 0 \\ 1 & 1 & 0 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 1 & 1 \\ 0 & 0 \end{bmatrix} = \begin{bmatrix} (1)(1) + (1)(1) + (0)(0) & (1)(1) + (1)(1) + (0)(0) \\ (1)(1) + (1)(1) + (0)(0) & (1)(1) + (1)(1) + (0)(0) \end{bmatrix} = \begin{bmatrix} 2 & 2 \\ 2 & 2 \end{bmatrix}$$

To find the eigenvalues, we solve the characteristic equation $\det(A^T A - \lambda I) = 0$:

$$\det \begin{bmatrix} 2 - \lambda & 2 \\ 2 & 2 - \lambda \end{bmatrix} = (2 - \lambda)^2 - 4 = 0$$

$$4 - 4\lambda + \lambda^2 - 4 = 0$$

$$\lambda^2 - 4\lambda = 0$$

$$\lambda(\lambda - 4) = 0$$

Therefore, the eigenvalues, ordered from largest to smallest, are:

$$\lambda_1 = 4, \quad \lambda_2 = 0$$

4.2 Find the singular values of A .

The singular values σ_i are the square roots of the eigenvalues of $A^T A$.

$$\sigma_1 = \sqrt{\lambda_1} = \sqrt{4} = 2$$

$$\sigma_2 = \sqrt{\lambda_2} = \sqrt{0} = 0$$

4.3 Find an orthogonal matrix V whose columns are right singular vectors.

We need to find the orthonormal eigenvectors for $A^T A$, which will form the columns of $V = [\mathbf{v}_1, \mathbf{v}_2]$.

For $\lambda_1 = 4$:

$$(A^T A - 4I)\mathbf{v}_1 = \begin{bmatrix} -2 & 2 \\ 2 & -2 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

This gives $-2x + 2y = 0 \implies x = y$. A corresponding unit vector is:

$$\mathbf{v}_1 = \begin{bmatrix} 1/\sqrt{2} \\ 1/\sqrt{2} \end{bmatrix}$$

For $\lambda_2 = 0$:

$$(A^T A - 0I)\mathbf{v}_2 = \begin{bmatrix} 2 & 2 \\ 2 & 2 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

This gives $2x + 2y = 0 \implies x = -y$. A corresponding unit vector is:

$$\mathbf{v}_2 = \begin{bmatrix} -1/\sqrt{2} \\ 1/\sqrt{2} \end{bmatrix}$$

Placing these as columns, our matrix V is:

$$V = \begin{bmatrix} \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \end{bmatrix}$$

4.4 Find an orthogonal matrix U and a 3×2 diagonal matrix Σ such that $A = U\Sigma V^T$

Find Σ : Since A is a 3×2 matrix, Σ must also be a 3×2 matrix. We place the singular values on the main diagonal and pad with zeros:

$$\Sigma = \begin{bmatrix} \sigma_1 & 0 \\ 0 & \sigma_2 \\ 0 & 0 \end{bmatrix} = \begin{bmatrix} 2 & 0 \\ 0 & 0 \\ 0 & 0 \end{bmatrix}$$

Find U : The columns of $U = [\mathbf{u}_1, \mathbf{u}_2, \mathbf{u}_3]$ are the left singular vectors. For non-zero singular values, we can find $\mathbf{u}_i$ using the formula $\mathbf{u}_i = \frac{1}{\sigma_i} A \mathbf{v}_i$.

For $\sigma_1 = 2$:

$$\mathbf{u}_1 = \frac{1}{2} \begin{bmatrix} 1 & 1 \\ 1 & 1 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} 1/\sqrt{2} \\ 1/\sqrt{2} \end{bmatrix} = \frac{1}{2} \begin{bmatrix} 2/\sqrt{2} \\ 2/\sqrt{2} \\ 0 \end{bmatrix} = \begin{bmatrix} 1/\sqrt{2} \\ 1/\sqrt{2} \\ 0 \end{bmatrix}$$

Because $\sigma_2 = 0$, we cannot use the formula to find $\mathbf{u}_2$. Instead, we must find vectors $\mathbf{u}_2$ and $\mathbf{u}_3$ such that $\mathbf{u}_1, \mathbf{u}_2, \mathbf{u}_3$ form an orthonormal basis for $\mathbb{R}^3$ (meaning they are unit vectors mutually orthogonal to each other).

To find $\mathbf{u}_2$ orthogonal to $\mathbf{u}_1$, we need a vector $\begin{bmatrix} x \\ y \\ z \end{bmatrix}$ such that $\frac{1}{\sqrt{2}}x + \frac{1}{\sqrt{2}}y + 0z = 0$, which means $x = -y$. Let's pick a normalized vector satisfying this:

$$\mathbf{u}_2 = \begin{bmatrix} -1/\sqrt{2} \\ 1/\sqrt{2} \\ 0 \end{bmatrix}$$

To find $\mathbf{u}_3$ orthogonal to both $\mathbf{u}_1$ and $\mathbf{u}_2$, we can take their cross product, or simply observe that the z-axis unit vector is orthogonal to the xy-plane vectors we just found:

$$\mathbf{u}_3 = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$$

Putting it all together, our orthogonal matrix U is:

$$U = \begin{bmatrix} \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} & 0 \\ \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

This yields the complete decomposition $A = U\Sigma V^T$.